Bayside Property Analysis - Assignment 3

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AD 571

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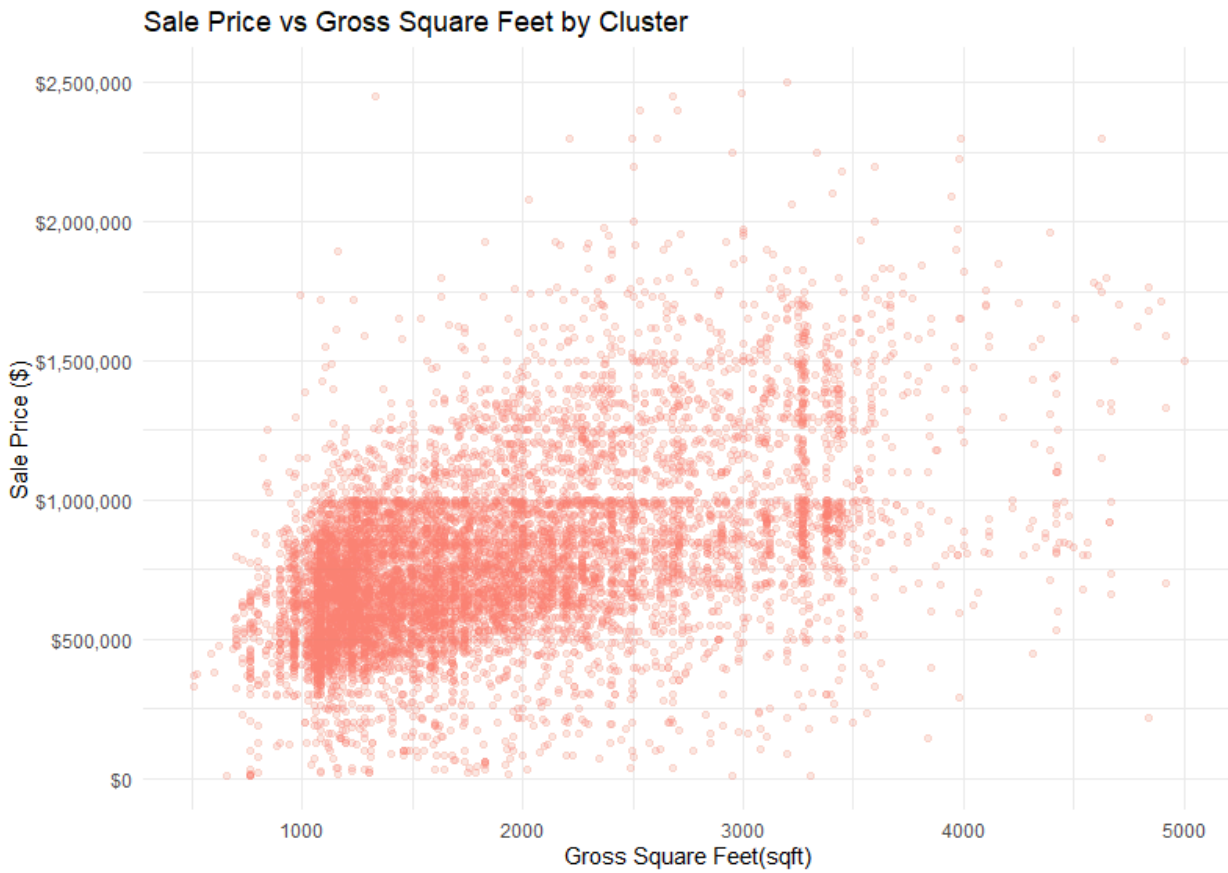
Initial Filtering:
$10,000 <= Price <= $2,500,000
0 < Gross Square Feet <= 5,000

Create variable Property Age at Sale → >= 0

1. The scatter plot below, between Sale Price ($) and Gross Square Feet (sqft), shows that there is a weak positive relationship between the two variables. This suggests that when the Gross Square Feet increases, so will the Price. However, due to the weak nature of this relationship caused by outliers and other variables, this may not always be the case.



2. Below is the Linear Regression model for predicting price using group square feet.

```
Call:
lm(formula = SALE_PRICE ~ GROSS_SQUARE_FEET, data = bayside_residential)

Residuals:
     Min       1Q   Median       3Q      Max 
-1171213  -168363   -26839   159087  1772348 

Coefficients:
                   Estimate Std. Error t value Pr(>|t|)    
(Intercept)       4.076e+05  7.801e+03   52.24   <2e-16 ***
GROSS_SQUARE_FEET 2.032e+02  3.867e+00   52.55   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 279600 on 8990 degrees of freedom
Multiple R-squared:  0.235,    Adjusted R-squared:  0.2349
F-statistic:  2762 on 1 and 8990 DF,  p-value: < 2.2e-16
```

3. Prediction using this linear regression model. This is solved in R, predicted price for 1500sqft below.

Predicted Price = $712,405.60

4. Data Cleaning

- Removed Commercial Units → not relevant as only using residential
- Removed Type of Property → not relevant as only using residential
- Removed Neighborhood Name → only looking at Bayside

Converted to factor

- Sale ID, Neighborhood ID, Zip Code, Borough ID, Apartment Number

Converted Remaining Character Variables into Factors to prepare for multi-linear regression.

5. Further Data Processing

- Removed Neighborhood ID, Borough ID due to having only 1 unique ID
- Removed Sale ID, Address, Apartment number due to having more than 50 unique ID

For the remaining Factors → order by most common unique values, keep only the top 6

6. Correlation Matrix

Sale Date, Sale Year removed as only need to look at Property Age at Sale. Keeping would be redundant.

```
RESIDENTIAL_UNITS GROSS_SQUARE_FEET  SALE_PRICE PROPERTY_AGE_AT_SALE
RESIDENTIAL_UNITS            1.0000000         0.5397640  0.26345479
-0.22140398
GROSS_SQUARE_FEET            0.5397640         1.0000000  0.48878588
-0.19491890
SALE_PRICE                   0.2634548         0.4887859  1.00000000
-0.04425126
PROPERTY_AGE_AT_SALE        -0.2214040        -0.1949189 -0.04425126
1.00000000
```

No correlations above 0.7, so not further processing required.

7. Linear regression after removing redundant variables. Redundant variables include sale date, sale year, year built, description and residential units. These were removed by the initial model, so I removed them in advance to show the final, cleaned model.

INITIAL MULTI-LINEAR MODEL

```
Call:
lm(formula = SALE_PRICE ~ ., data = bayside_residential)

Residuals:
     Min       1Q   Median       3Q      Max 
-1203950   -86426     6098    94897  1648068 

Coefficients: (7 not defined because of singularities)
                                                 Estimate Std. Error t value 
Pr(>|t|)
(Intercept)                                     1.748e+07  1.654e+07   1.057 
0.2907
BUILDING_CLASS_FINAL_ROLLA1                     1.390e+04  9.001e+03   1.544 
0.1225
BUILDING_CLASS_FINAL_ROLLA2                     5.546e+03  9.765e+03   0.568 
0.5701
BUILDING_CLASS_FINAL_ROLLA5                    -1.109e+05  9.008e+03 -12.305 
< 2e-16 ***
BUILDING_CLASS_FINAL_ROLLB1                     2.290e+04  1.191e+04   1.922 
0.0546 .
BUILDING_CLASS_FINAL_ROLLB2                    -2.299e+04  1.208e+04  -1.903 
0.0571 .
ZIP_CODE11358                                  -7.962e+03  2.607e+04  -0.305 
0.7600
ZIP_CODE11360                                   6.974e+04  2.387e+04   2.921 
0.0035 **
ZIP_CODE11361                                   4.597e+03  2.332e+04   0.197 
0.8437
ZIP_CODE11364                                   9.329e+03  2.346e+04   0.398 
0.6909
ZIP_CODE11693                                   8.947e+03  9.424e+04   0.095 
0.9244
RESIDENTIAL_UNITS                                      NA         NA      NA 
NA
GROSS_SQUARE_FEET                               1.839e+02  4.805e+00  38.270 
< 2e-16 ***
YEAR_BUILT                                      3.976e+02  6.508e+01   6.110 
1.05e-09 ***
SALE_DATE                                       9.687e+01  2.297e+01   4.217 
2.50e-05 ***
DESCRIPTIONONE FAMILY ATTACHED OR SEMI-DETACHED         NA         NA      NA 
NA
DESCRIPTIONONE STORY - PERMANENT LIVING QUARTER         NA         NA      NA 
NA
DESCRIPTIONTWO FAMILY BRICK                             NA         NA      NA 
NA
DESCRIPTIONTWO FAMILY FRAME                             NA         NA      NA 
NA
DESCRIPTIONTWO STORIES - DETACHED SM OR MID             NA         NA      NA 
NA
```

```
SALE_YEAR                                      -9.604e+03  8.398e+03  -1.144
0.2528
PROPERTY_AGE_AT_SALE                                   NA         NA      NA
NA
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 203400 on 7488 degrees of freedom
Multiple R-squared:  0.564,    Adjusted R-squared:  0.5632
F-statistic:   692 on 14 and 7488 DF,  p-value: < 2.2e-16
```

CLEANED MULTI-LINEAR MODEL → lower R-Squared Value, but removed the redundant variables.

```
Call:
lm(formula = SALE_PRICE ~ ., data = bayside_residential)

Residuals:
     Min       1Q   Median       3Q      Max
-1069718  -152475   -28653   153055  1744826

Coefficients:
                             Estimate Std. Error t value Pr(>|t|)
(Intercept)                 4.590e+05  3.217e+04  14.269  < 2e-16 ***
BUILDING_CLASS_FINAL_ROLLA1  7.109e+03  1.149e+04   0.619  0.53624
BUILDING_CLASS_FINAL_ROLLA2  1.141e+04  1.247e+04   0.915  0.36012
BUILDING_CLASS_FINAL_ROLLA5 -1.247e+05  1.150e+04 -10.839  < 2e-16 ***
BUILDING_CLASS_FINAL_ROLLB1  2.890e+04  1.521e+04   1.900  0.05743 .
BUILDING_CLASS_FINAL_ROLLB2 -1.017e+04  1.543e+04  -0.659  0.50969
ZIP_CODE11358              -3.912e+04  3.329e+04  -1.175  0.23993
ZIP_CODE11360               8.086e+04  3.049e+04   2.652  0.00802 **
ZIP_CODE11361              -1.701e+04  2.978e+04  -0.571  0.56773
ZIP_CODE11364              -1.469e+04  2.996e+04  -0.490  0.62399
ZIP_CODE11693               1.867e+05  1.203e+05   1.552  0.12072
GROSS_SQUARE_FEET           1.714e+02  6.132e+00  27.949  < 2e-16 ***
PROPERTY_AGE_AT_SALE        4.597e+02  8.157e+01   5.636 1.81e-08 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 259800 on 7490 degrees of freedom
Multiple R-squared:  0.2886,   Adjusted R-squared:  0.2875
F-statistic: 253.3 on 12 and 7490 DF,  p-value: < 2.2e-16
```

Statistically Insignificant Variables

- Building Class (A1, A2, B1, B2)
- Zip Code (11358, 11361, 11364, 11693)

8. Prediction using multi-linear regression model

Gross Square Feet = 1589
Property Age at Sale = 61
Zip Code = 11361
Building Class Final Roll = A1

Predicted Price = $749,457.30

9. Predictions of Biggest Bargains and Most Overpriced

```
[1] "Biggest Bargains:"
  SALE_PRICE PREDICTED_PRICE PRICE_DIFF
       <dbl>          <dbl>      <dbl>
1      10000        1079718.  -1069718.
2      35000         994048.   -959048.
3     178856        1137350.   -958494.
```

```
[1] "Most Overpriced:"
  SALE_PRICE PREDICTED_PRICE PRICE_DIFF
       <dbl>          <dbl>      <dbl>
1    2450000         705174.   1744826.
2    2300000         834124.   1465876.
3    2460000        1012895.   1447105.
```

When considering how to account for the discrepancies, there are a number of important factors to consider…

1. Model Limitations: the model may not account for all of the key factors that will influence sale price. Additional factors may include the property condition, recent renovations, specific location, or luxury features. These may influence the price above and below the prediction.
2. Market Anomalies: these properties may have unique circumstances around their sales, which may include distressed sales, family transfers or extreme bidding wars which lead to drastic changes in their sale price. These factors do not align with standard market logic but are not impossible.
3. Data Issues: The data may have been incorrectly recorded or input, which means that the sale price recorded has the possibility of not being accurate.

These three factors offer insight to the potential differences between predicted and actual price. From our analysis, they are anomalies but there is likely to be an explanation with more information as to why the prediction varied from the actual value.